UNBC UNIVERSITY OF NORTHERN BRITISH COLUMBIA

Ken A. Otter, Professor & Department Chair
Department of Ecosystem Science & Management

Application – Ms. Yi-Chin (Sunny) Tseng,

I would like to offer my recommendation for Yi-Chin (Sunny) Tseng's application for the **Pairi Daiza Foundation Grant for Biodiversity and Conservation Research**.

Ms. Tseng has been working in my research group since 2021 on her PhD research. Ms. Tseng came highly recommended from her MSc in biostatistics and forest ecology at UBC. Ms. Tseng's background in both R and Python programming was somewhat unique, especially when coupled with her keen interest in audio recording and bird song. With the recent advances in Autonomous Recording Units (ARU) for avian censusing, I saw the potential for someone like Ms. Tseng to create major advances in the field of auto-identification of species and management of the massive datasets these could provide. Within the past ten years, ARU have been systematically replacing traditional "point-counts" used to identify the presence of different bird communities in forested landscapes. These passive monitoring units allow researchers to deploy recording units over extended periods and collect data simultaneously from multiple sites to compare bird communities between areas under different harvesting management. Their primary downfall has been the massive amount of personnel time required to transcribe and manage the massive amount of data these systems can provide.

Ms. Tseng had already had significant experience recording and analyzing bird sounds when she first contacted me about beginning doctoral work. Between her Masters and starting at UNBC, she returned to her native Taiwan to work with the Endemic Species Institute to deploy and test ARU in biodiversity monitoring in remote parts of the country. She designed a similar sampling protocol with her PhD work at the John Prince Research Forest in northern British Columbia, Canada – this is a 16,000ha community forest tenure co-operated between UNBC and the Tl'atz'en First Nation. For her PhD dataset, she collected three years of ARU surveys from over 60 locations within and around the Research Forest (2020-2022). The start of her PhD research also coincided with the release of the BirdNET Deep Learning Algorithm for species recognition of birds from audio recordings, and she quickly began to test the capacity of this system for automating species detections from ARU recordings. She recently published the result of this work developing the protocols researchers can use for calibrating the precision of BirdNET detections to ensure high data accuracy – the originators of BirdNET were so impressed with her work that they invited her to undertake a three-month Internship (completed in the summer of 2025) to help create the interface between BirdNETs python operating language and R. This has resulted in the R package that Ms. Tseng created (BirdNETtools), which will expand the utility of this system to a far greater community of potential users in avian ecology research.

Ms. Tseng has already published two of the four chapters that will comprise her PhD thesis, will be submitting another of these chapters for publication in Nov 2025, and is well on her way to completing the final chapter of her work. I anticipated she will have successfully defended her doctoral work by April 2026, leaving her open to conduct independent field work in spring/summer 2026. She has already proven her ability to balance thesis completion with pursuing outside projects; this included not only the three-month Internship with BirdNET from May to July 2025, but while in Europe, Ms. Tseng implemented a pilot program to deploy ARU in Lithuania to begin documenting the diversity in their avifauna, and I believe that the proposed extension of this field work will allow her to help establish the

Dept of Ecosystem Science & Management
University of Northern British Columbia
3333 University Way, Prince George, BC V2N 4Z9
Phone: 250-960-5919 • Fax: 250-960-5539 • email: ken.otter@unbc.ca

country's first long-term passive acoustic monitoring program to document migratory birds. This project will help establish her as a independent researcher and will further establish her commitment and contributions in avian conservation.

I strongly recommend Sunny for the Pairi Daiza Foundation Grant.

Sincerely,

Ken A. Otter

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